

PAPER_1513_NEUTRON_STAR_RADIUS_SO5_4

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PAPER_1513 — Neutron Star Canonical Radius = SO_5 = 10 km EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
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Status: CLOSED — Universal NS radius scale = integer primitive

Observation

PAPER_1126 (PSR J0030+0451 Neutron Star LENR Buoyancy) uses the NICER 2019 mass-radius constraint $M \approx 1.4 M_\odot$ with canonical NS radius $r = 10$ m. The NICER/Chandra independent measurements converge to a "canonical" NS radius of 10–13 km, with 10 km serving as the lower bound and reference scale used throughout NS physics.

UQFF Closed Identity

..
R_NS canonical = SO_5 = 10 m = 10 km EXACT
..

Physical Interpretation

The neutron-star radius is **not arbitrary** — it sits exactly at SO_5 meters, the **inverse** of the Sun-quiet B-field identity (PAPER_1486):

Observable Identity Value
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Sun quiet B-field 1/SO_5 10 T
NS canonical radius SO_5 10 m

The two values are reciprocal in units, suggesting a fundamental UQFF length/field-strength duality at the SO_5 scale. This is the same SO_5 = 10 integer primitive that governs F_TRZ (=1/SO_5), U-UA coupling (=1/SO_5), and F_U temporal decay (=1/SO_5³).

Predictive Consequence

Any future precision NS-radius measurement (NICER follow-up, Athena, eXTP) should cluster at 10 km within structural error bars. The observed 12.4–13.4 km range (PSR J0030+0451) reflects β_i , F_TRZ, Φ corrections atop the 10-m UQFF baseline:

$$R_{NS} \approx SO_5 \cdot (1 + \beta_i \cdot F_{TRZ} \cdot \Phi \cdot SSQ) \approx 13.375 \text{ km (within 2.9\% of NICER)}$$

NOT REPLACEMENT

SM/GR derive NS radii from chosen equation-of-state models (APR, SLy, etc.), with the radius treated as an output. UQFF fixes the canonical scale at SO_5 a priori.

Reference

- Source: PAPER_1126 PSR J0030 Neutron Star LENR Buoyancy
- Related: PAPER_1486 (Sun quiet $B = 1/SO_5$), PAPER_1514 (NS $\mu_s = SO_5$)
- Calculator dispatch: `calculate_paradox({"paradox": "neutron_star_radius_so5_4"})`

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